

Inter Pipeline Propylene Ltd.
Propane Dehydrogenation Project



10-P-4400XA/B Series Pump Dynamic Analysis
PDH-00-ST-CAL-0004



This document has been revised as indicated below. Please destroy all previous revisions.

Rev	Date	Revision Description	PREP	CHKD	APP'D	MANAGEMENT	
						Fluor	IPPL
0	7-NOV-2017	Issued for Information – Project 1202	RPM	SC	-	-	-

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PROJECT: Propane Dehydrogenation Project **Contract No.:** A6KV
Location: Strathcona County, Alberta **Date:** 28-FEB-2017
Title: 10-P-4400XA/B Series Pump Dynamic Analysis **Rev:** 0
Calculation No.: PDH-00-ST-CAL-0004 **Check Level:** 2B
By: R. Mitchell **Chk'd:** S. Chen

Drawing Numbers - Titles:

Stormwater Run-Off Pond Discharge Sump Plan – PDH-CV01-00005

Design Specifications:

PDH-00-ST-SPC-0001 STRUCTURAL ENGINEERING DESIGN CRITERIA

Approvals for First Issue

	Name (Print)	Signature	Date (day/mo/yr)
Originator	Robert Mitchell	Robert Mitchell	02/MAR/2017
Checker	Skyler Chen	Skyler Chen	06/MAR/2017
Engineer of Record	Skyler Chen	Skyler Chen	06/MAR/2017

Record of Revisions

Rev. No.	Reason for Revision (Only for changes after first issue)	Date (day/mo/yr)	Engineer	Checked By

PROJECT:	Propane Dehydrogenation Project	Contract No.:	A6KV
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TABLE OF CONTENTS

<u>SECTION</u>	<u>DESCRIPTION</u>	<u>PAGE</u>
1.0	DESIGN SUMMARY	4
2.0	UNCERTAINTIES AND RISKS.....	8
3.0	DESIGN	10
4.0	DYNA 5 ANALYSIS	13
5.0	SAP2000 ANALYSIS	19
A	APPENDIX A: VENDOR INFORMATION.....	24

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Section 1.0: Design Summary

PROJECT:	Propane Dehydrogenation Project	Contract No.:	A6KV
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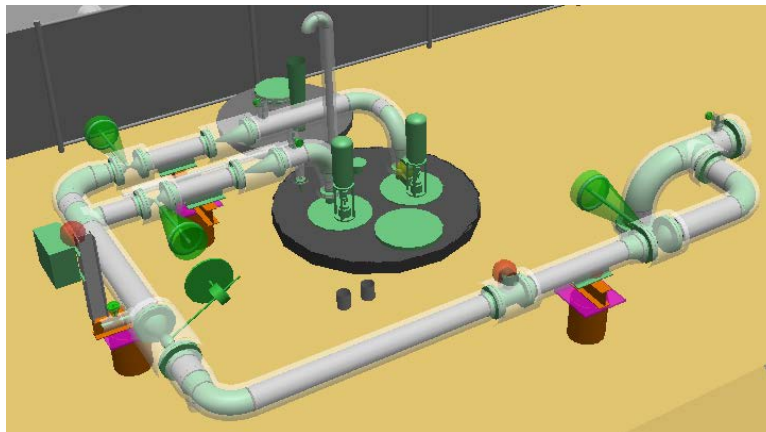
1.0 DESIGN SUMMARY

1.1 Purpose of Calculation

The calculation is intended to document design basis, design criteria, and calculation for the dynamic analysis of pumps 10-P-44001A/B & 10-P-44003A/B in design area 00AC (see Appendix A for vendor data).

1.2 Structure Layout

S3D model snapshots (as of 23-Feb-2017) shown below.



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1.3 Method of Analysis

Manual calculations in accordance with the design criteria listed in Section 1.4 below, based on vendor data contained in Appendix A.

1.4 Design Criteria

1.4.1 General

Refer to Drawings A6KV-4-0605-01-00056 & A6KV-4-0605-01-00132 in Appendix A.

1.4.2 Governing Codes

NBC 2005	National Building Code of Canada 2005
ABC 2006	Alberta Building Code 2006
CAN/CSA-S16	Limit States Design of Steel Structures

1.4.3 Specifications

PDH-00-ST-SPC-0001	STRUCTURAL ENGINEERING DESIGN CRITERIA
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1.4.4 Strength and Serviceability Requirements

CAN/CSA-S16	Limit States Design of Steel Structures
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1.4.5 Materials

N/A

1.4.6 References

IPPL PDH Project Job Data Book:
Jdb 2.04: Pile Design

1.4.7 Equipment List

Equip Tag	Drawing / Doc No	Revision	Description
10-P-44001A/B	A6KV-4-0605-01-00132	1	Pump GA Drawing
10-P-44003A/B	A6KV-4-0605-01-00056	1	Pump GA Drawing

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1.5 Software

1.5.1 Electronic Files

File Name	Description	Directory
PDH-00-ST-CAL-0004.pdf	The Calculation File	\\fdnet.com\ca_projects\CA_A6KV\215_STRUCTURAL\215E_ENGINEERING_DATA\215E_014_R_Proj_Calculations_Sketches\PDH-00-ST-CAL-0004_00AC-CC- Dynamic Analysis of 10-P-4400XA-B Series Pumps
PDH-00-ST-CAL-0004 – Cover Sheet.doc	Calculation Cover Sheet	
PDH-00-ST-CAL-0004 - Calculation Spreadsheet.xls	Calculation Spreadsheet	
PDH-00-ST-CAL-0004.dy5	The Dyna5 Model File	
PDH-00-ST-CAL-0004.vts	The Dyna5 VTS File	
PDH-00-ST-CAL-0004.txt	The Dyna5 Output File	
PDH-00-ST-CAL-0004.sdb	The SAP2000 Model File	

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Section 2.0: Uncertainties and Risks

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2.0 UNCERTAINTIES AND RISKS

Uncertainty	Mitigation	Remaining Risk	Comment	Resolution

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Section 3.0: Design

PROJECT:	Propane Dehydrogenation Project	Contract No.:	A6KV
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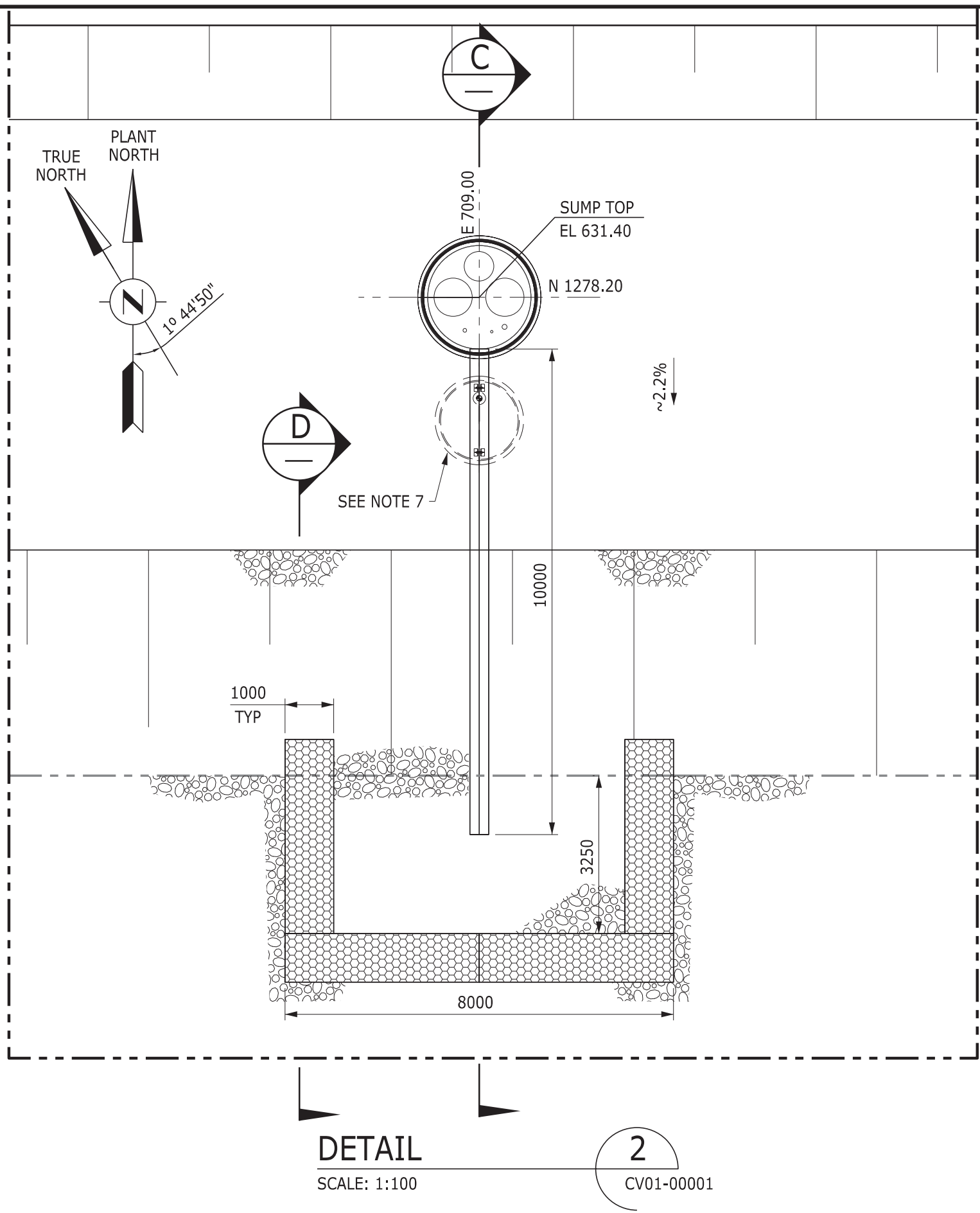
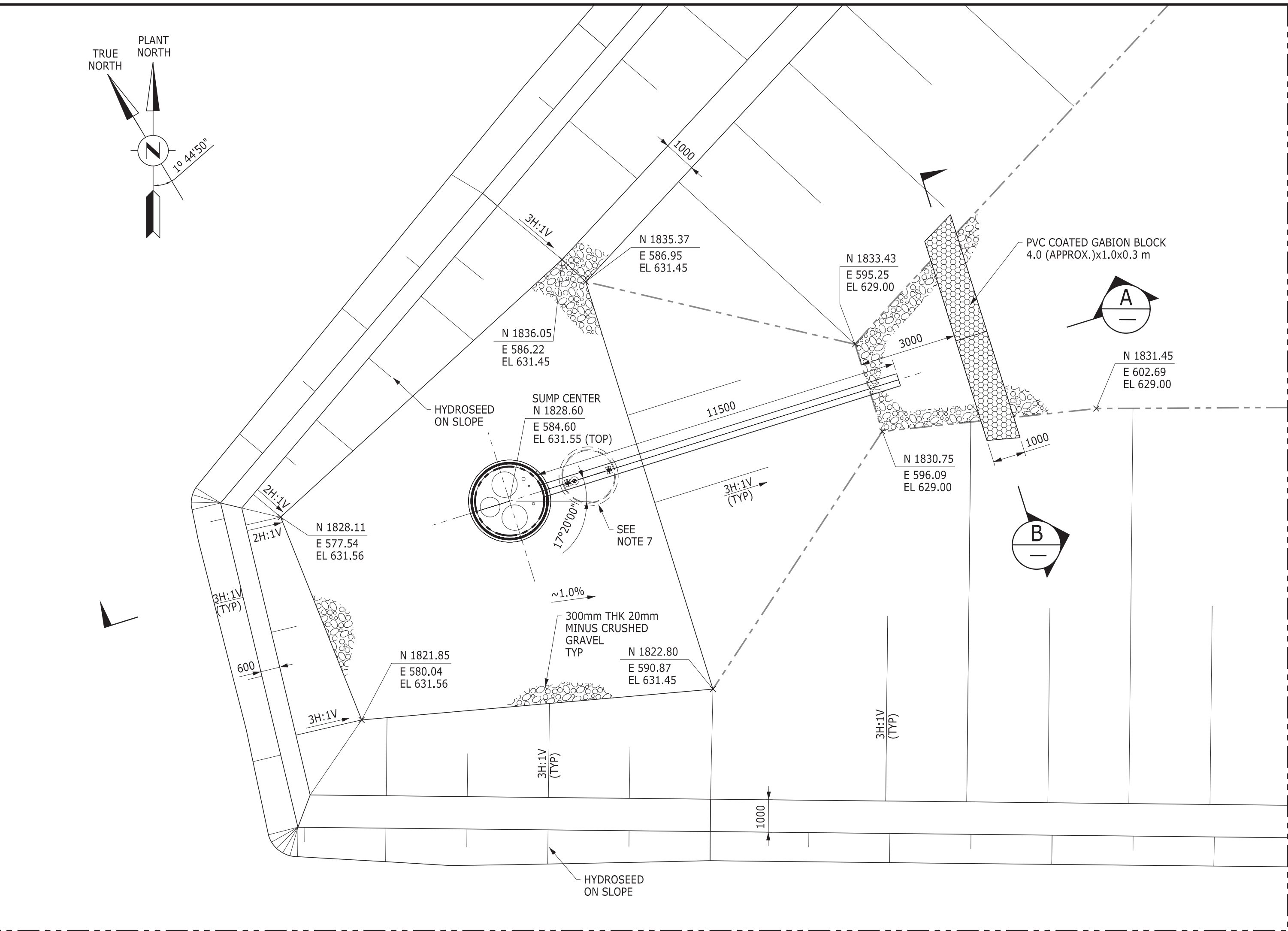
3.0 DESIGN

3.1 Basis

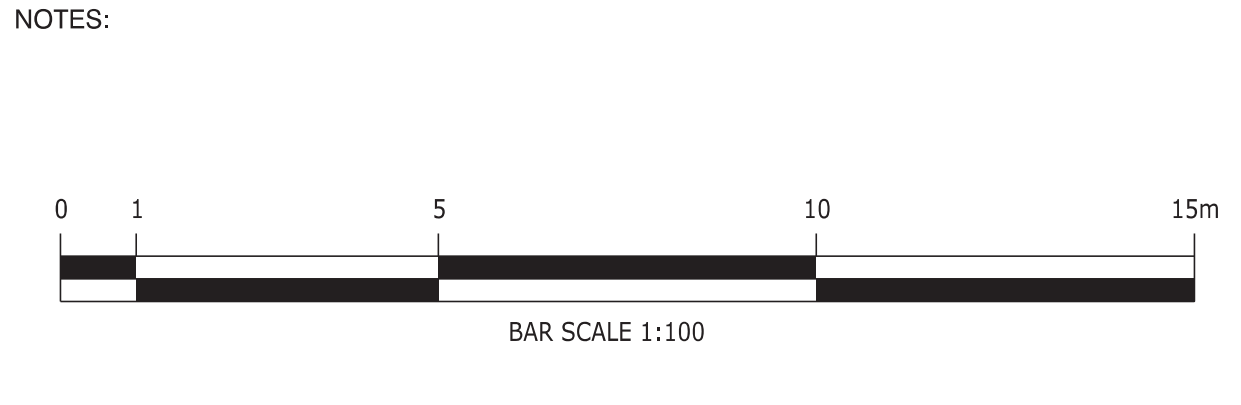
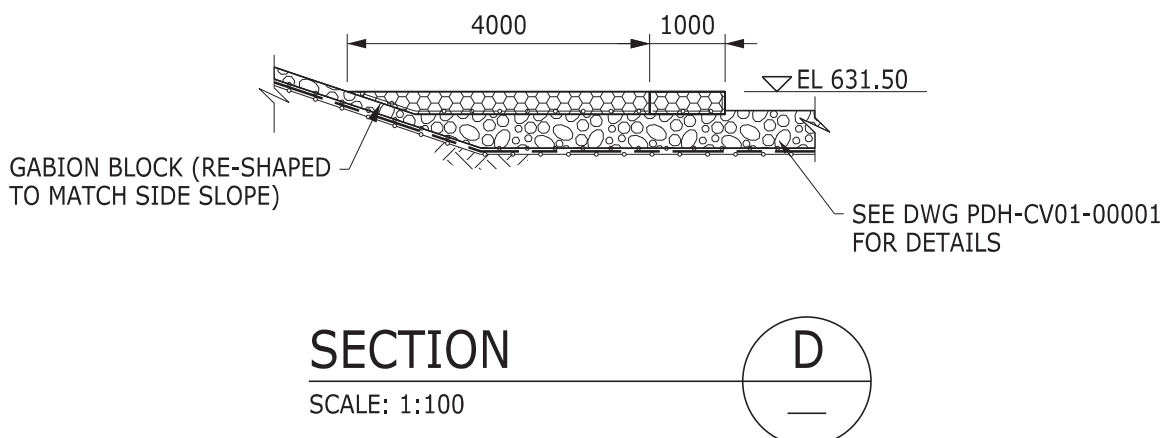
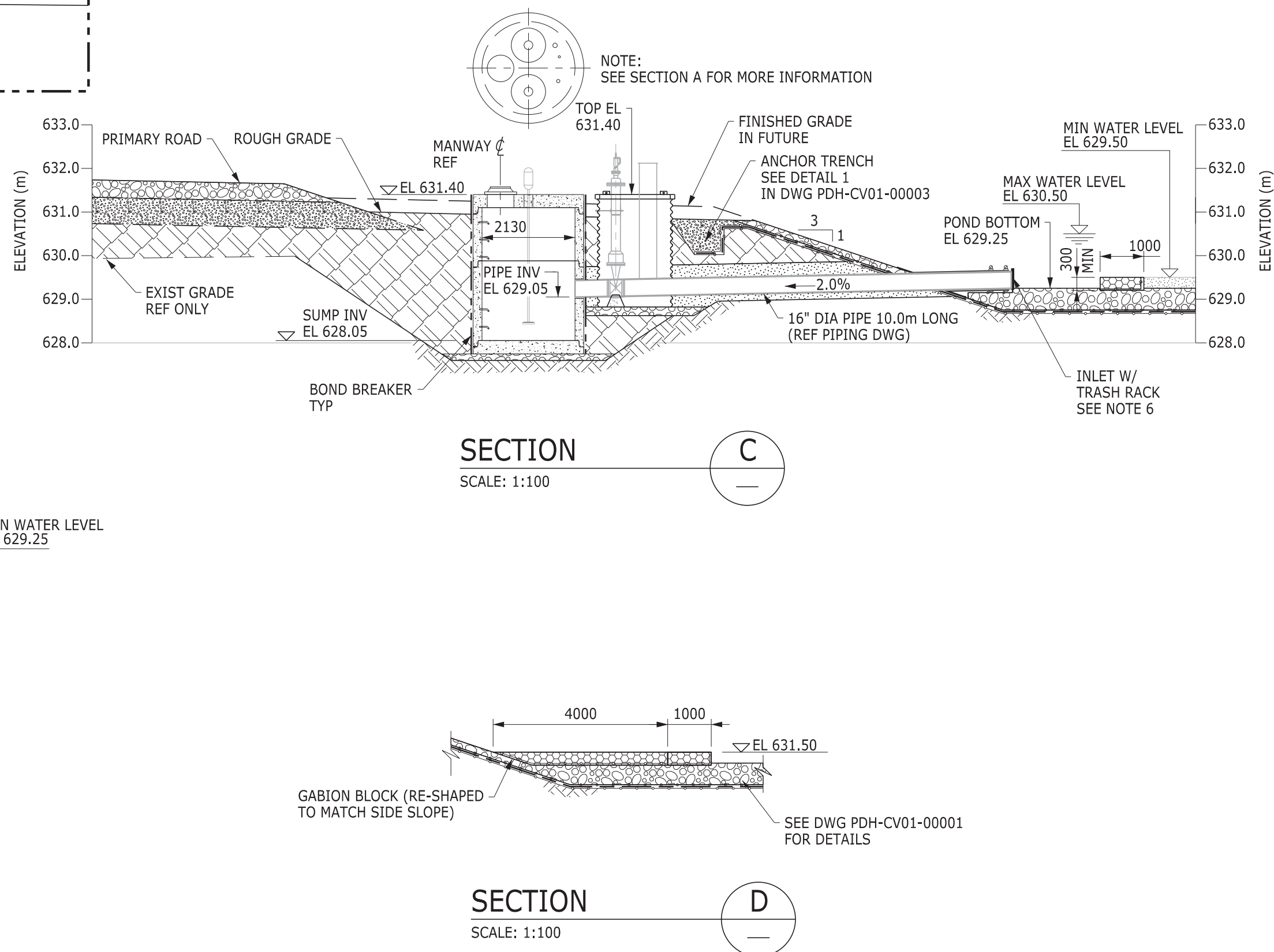
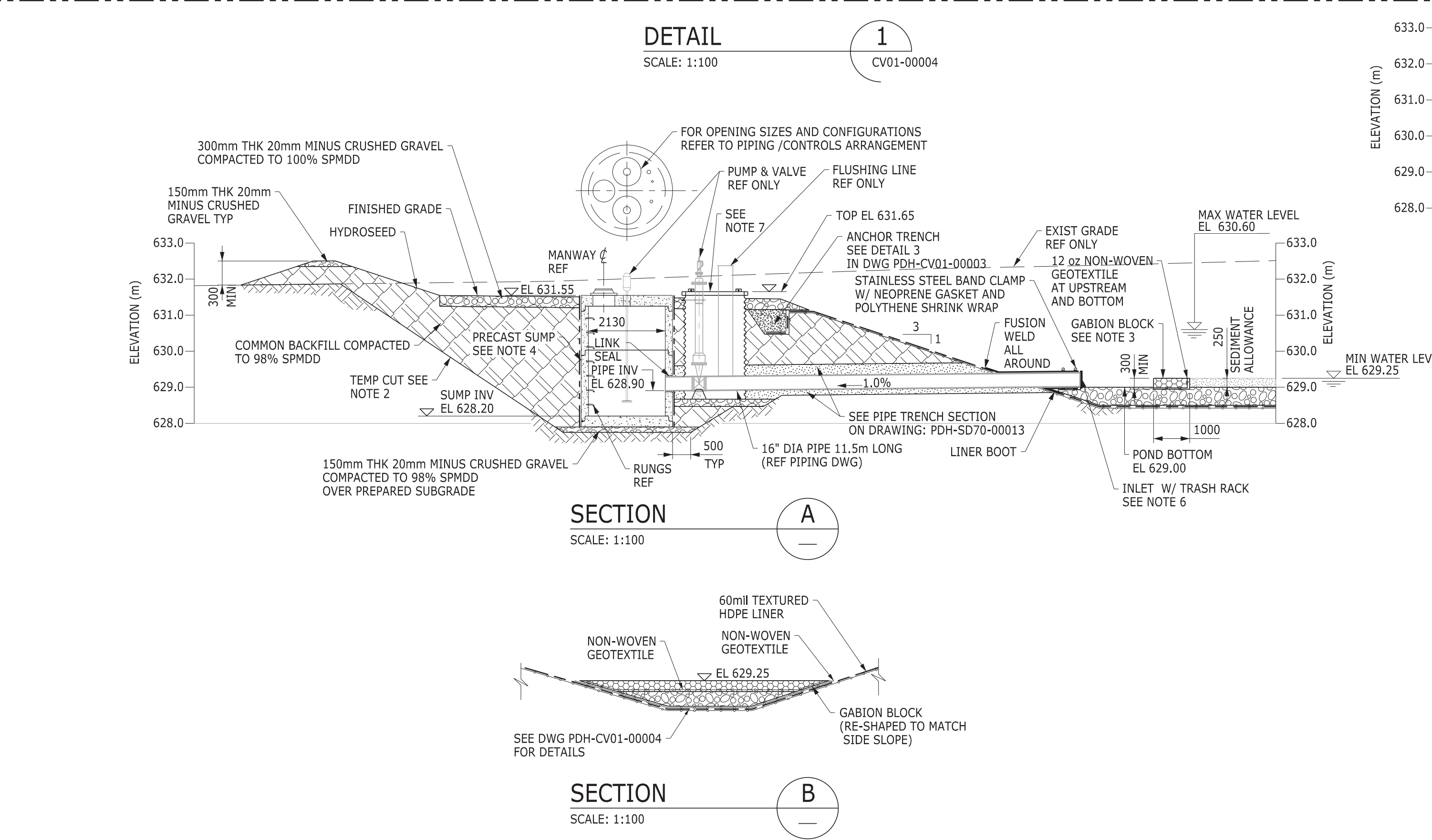
A dynamic analysis of pumps 10-P-44001A/B & 10-P-44003A/B is performed to ensure that the pumps will not operate at a resonant frequency that will cause excessive vibrations which will damage the equipment over time. In this situation human factors are not a consideration for the dynamic analysis as the pumps are located in remote areas.

3.2 Method

The pumps sit on top of a precast catch basin provided by Lafarge (See drawing in Appendix A). The Sump lid will have openings for a manhole, two pumps and minor piping. The catch basin (excluding the cover) was modeled in Dyna 5 as a concrete foundation to determine the dynamic properties. The cover was modeled in SAP2000 and the catch basin was modeled as a pile using the dynamic properties determined with Dyna 5. Finally the pump was added to the model and the analysis was run to determine the resonant frequencies.



- NOTES:**
- FOR GENERAL NOTES SEE DRAWING PDH-CV01-00004.
 - ALL CUT SLOPES SHALL BE IN ACCORDANCE WITH THE ALBERTA OCCUPATIONAL HEALTH & SAFETY CODE AND AS PER GEOTECHNICAL RECOMMENDATIONS.
 - GABION BLOCK SHALL BE ARMTEC PVC COATED STEEL WIRE 12 GAUGE OR EQUIVALENT.
 - LAFARGE PRECAST SUMP OR EQUIVALENT SHALL BE INSTALLED AS PER MANUFACTURER'S INSTALLATION DETAILS AND PROCEDURES. WRAP THE SUMP ALL AROUND WITH A BOND BREAKER. FOR MANHOLE VENT AND COVER DETAILS SEE PIPING DAWRING.
 - NON-WOVEN GEOTEXTILE SHOWN IN DRAWING IS NILEX 4512 OR EQUIVALENT.
 - END CAP PIPE WITH WIRE MESH TO PREVENT RODENT ENTRY, SECURE TO PIPE WITH STAINLESS STEEL BAND CLAMPS AS APPROVED BY FIELD ENGINEER.
 - 1600mm DIA CSP MANHOLE FOR VALVE, SEE DRAWING PDH-SD70-00012. NO SAND FILLING INSIDE MANHOLE. FOR VALVE LOCATION SEE PIPING DRAWING.



REFERENCE DWG.'s			REVISIONS			
2	PDH STORMWATER POND - PLAN AND SECTION	PDH-CV01-00001				
1	ADMIN/RAILYARD STORMWATER POND - PLAN AND SECTION	PDH-CV01-00004	0	10-NOV-2016	ISSUED FOR CONSTRUCTION - PROJECT 1202	
NO.	DESCRIPTION	DWG. NO.	NO.	DATE	DESCRIPTION	

PERMIT STAMP		ENGINEER STAMP	
PERMIT TO PRACTICE FLUOR CANADA LTD.			
Signature			
Date			
PERMIT NUMBER: P 0778			
The Association of Professional Engineers, Geologists and Geophysicists of Alberta			

WILLIAMS CANADA PROPYLENE ULC			
ALBERTA PROPANE DEHYDROGENATION PLANT STORMWATER RUN-OFF POND DISCHARGE SUMP PLAN, SECTION AND DETAIL			
STRATHCONA COUNTY		ALBERTA	
DRAWN BY	S. WANG	SCALE: 1:100	A.F.E.:
CHECKED BY	V. MYANA	DRAWING NO.	
PROJECT ENG.	B. LI	PDH-CV01-00005	
APPROVED BY	G. GERALD		
DATE ISSUED	13-OCT-2016	LEGACY DRAWING NO.:	

EMP NO. 100046101AAEW, 1000AC101AAEW, 1000AC101AAEN

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Section 4.0: Dyna 5 Analysis

Section 4.0 - Dyna5 Analysis

VTS&3D Input

Rectangular Blocks

Corner_X	Corner_Y	Corner_Z	Length_X	Width_Y	Height_Z	Density
N/A	N/A	N/A	N/A	N/A	N/A	N/A

Cylindrical Blocks

Center_X	Center_Y	Center_Z	Radius	Height	Density	Orientation
0	0	-1.8265	1.268	3.653	2400	3
0	0	-1.824	1.065	3.048	-2400	3

Lumped Masses

X	Y	Z	Mass	IXX	IYY	IZZ	IXY	IXZ	IYZ
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

CBC

Base			
Radius	Center_X	Center_Y	Center_Z
1.268	0	0	-3.653

RBC

Base				
X_Length	Y_Length	Center_X	Center_Y	Center_Z
2.536	2.536	0	0	-3.653

Forces and Moments

X	Y	Z	Fx	Fy	Fz	Mx	My	Mz
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Total Mass = 18218.2

Center of Mass X-Coord. = 0

Center of Mass Y-Coord. = 0

Center of Mass Z-Coord. = -1.830077

Mass Moment of Inertial IXX = 39474.19

Mass Moment of Inertial IYY = 39474.19

Mass Moment of Inertial IZZ = 20818.26

Mass Moment of Inertial IXY = 0

Mass Moment of Inertial IXZ = 0

Mass Moment of Inertial IYZ = 0

Total Force in X-Direction = 0

Total Force in Y-Direction = 0

Total Force in Z-Direction = 0

Total Moment about X-Direction = 0

Total Moment about Y-Direction = 0

Total Moment about Z-Direction = 0

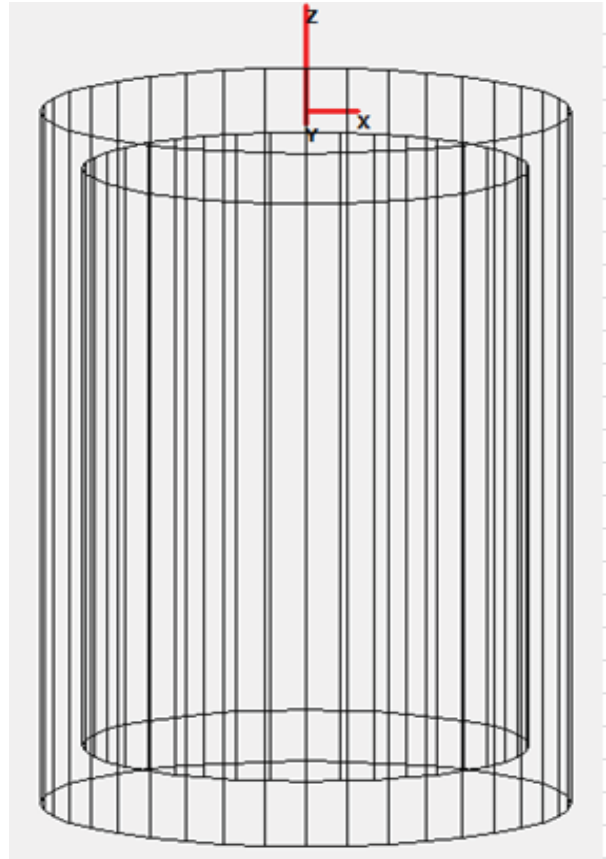
Length about X-Direction = 2.536

Length about Y-Direction = 2.536

Base center X-Direction = 0

Base center Y-Direction = 0

Base center Z-Direction = -1.83007717132568



Dyna5 Input

Setting Summary

Plot Height

☒ 51 Lines

☐ 105 Lines

Plot Width

☐ 70 Characters

☒ 120 Characters

Units

☐ Imperial Units (lb, slug, ft, g=32.2)

☒ SI Units (N, kg, m, g=9.81)

☐ Other

Frequency Units

☐ Rad/Sec

☒ Hertz

☐ RPM

Output Flags

☒ Echo Input Data

☐ Dimensionless Results

☒ Stiffness/Damping Matrices

☐ Output Draft Plots

Analysis Constants

Damping Safety Factor:

Gravity Acceleration:

Footing Base Dimensions and Data

Footing Mass & Inertia

Total Mass:

Ixx:

Iyy:

Izz:

Ixy:

Ixz:

Iyz:

Dimensions of Base

Length Lx:

Length Ly:

Coord. of Base Center

Xc:

Yc:

Zc:

Coord. of CG

X:

Y:

Z:

ELEVATION

PLAN

Soil Layers:

Layer 1:
Clay Till Fill

Layer 2:
Sand/Clay

Data for Cap/Footing Side Layers

Edit

Side Layers Properties

☐ Weak Zone

CANCEL OK

	Side Layer Thickness	Side Layer Shear Wave Velocity	Side Layer Unit Wt.	Poisson's Ratio	Damping
1	3.048	118.4	21000	0.35	0.04
2	0.305	99	20000	0.35	0.04
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					

Side Layer Elements

Properties of Soil Below the Found...

Shear Wave Velocity: 100

Soil Unit Weight: 20000

Poisson's Ratio: 0.35

Material Damping: 0.04

OK

CANCEL

Output Dynamic Properties:

FREQUENCY - 30.000 Hertz

STIFFNESS CONSTANTS (K) *****

HORIZONTAL TRANSLATION (X) ...	4.790E+08	N/m
HORIZONTAL TRANSLATION (Y) ...	4.790E+08	N/m
VERTICAL TRANSLATION (Y)	3.857E+08	N/m
ROTATION ABOUT (X)	5.331E+09	N.m/rad
ROTATION ABOUT (Y)	5.331E+09	N.m/rad
TORSION ABOUT (Z)	2.162E+09	N.m/rad
CROSS-STIFFNESS (YZ PLANE)	-1.434E+09	N/rad
CROSS-STIFFNESS (XZ PLANE)	1.434E+09	N/rad

DAMPING CONSTANTS (C) *****

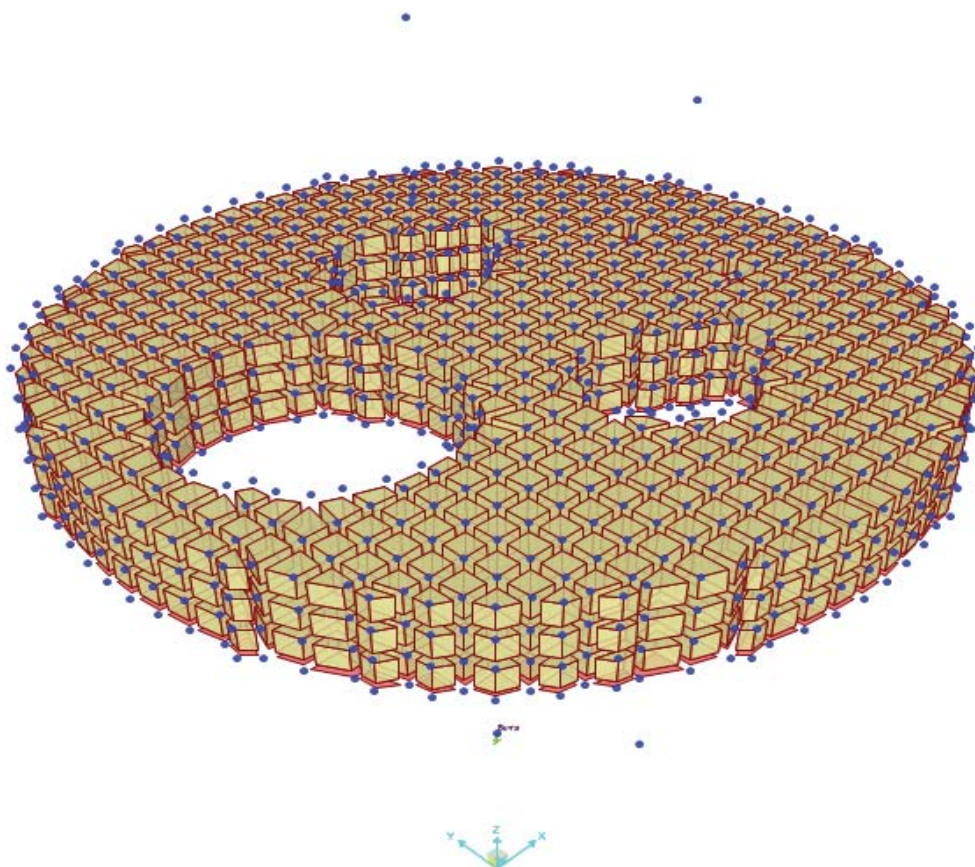
HORIZONTAL TRANSLATION (X) ...	6.566E+06	N/m/s
HORIZONTAL TRANSLATION (Y) ...	6.566E+06	N/m/s
VERTICAL TRANSLATION (Y)	5.127E+06	N/m/s
ROTATION ABOUT (X)	8.330E+07	N.m/rad/s
ROTATION ABOUT (Y)	8.330E+07	N.m/rad/s
TORSION ABOUT (Z)	8.273E+06	N.m/rad/s
CROSS-DAMPING (YZ PLANE)	-2.173E+07	N/rad/s
CROSS-DAMPING (XZ PLANE)	2.173E+07	N/rad/s

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Calculation No.:	PDH-00-ST-CAL-0004	Check Level:	2B
By:	R. Mitchell	Chk'd:	S. Chen

Section 5.0: SAP2000 Analysis

Section 5.0 - SAP 2000 Analysis

Model: The Sump Lid was modeled using elements approximately 100mm cubed.



10-P-44001A/B Pumps:

Component	Weight (kg)	Weight (kN)
Pump	471	4.62
Coupling	6	0.06
Motor	272	2.67
Total	749	7.35

Frequency: 1800 RPM / 30Hz

Power: 50 HP

10-P-44003A/B Pumps:

Component	Weight (kg)	Weight (kN)
Pump	444	4.36
Coupling	5	0.05
Motor	159	1.56
Total	608	5.96

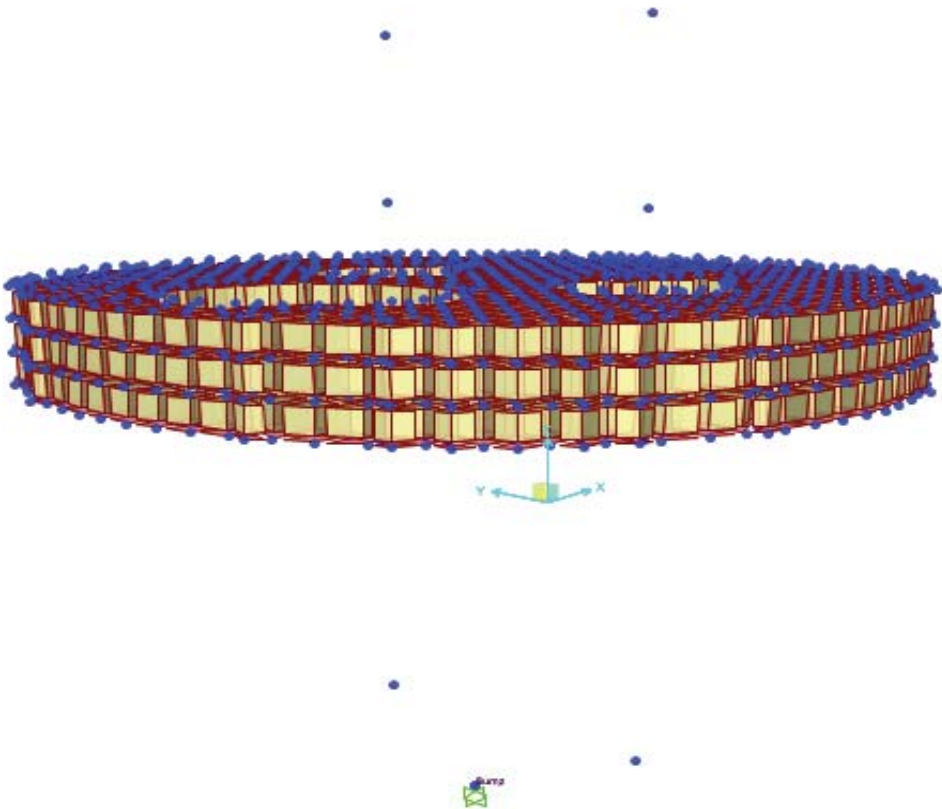
Frequency: 1800 RPM / 30Hz

Power: 25 HP

Only the 10-P-44001A/B pumps will be analysed, the 10-P-44001A/B pumps are lighter and will therefore be Ok by inspection as long as the 10-P-44003A/B pumps are adequate.

DATE:	02-Mar-2017
CONT. NO.	A6KV
BY:	RPM
CHK'D BY:	SC
SHEET NO:	
CALC NO.:	PDH-00-ST-CAL-0004

10-P-4400XA/B Series Pump Dynamic Analysis



<= Motor Mass

<= Coupling Mass

<= Pump Mass

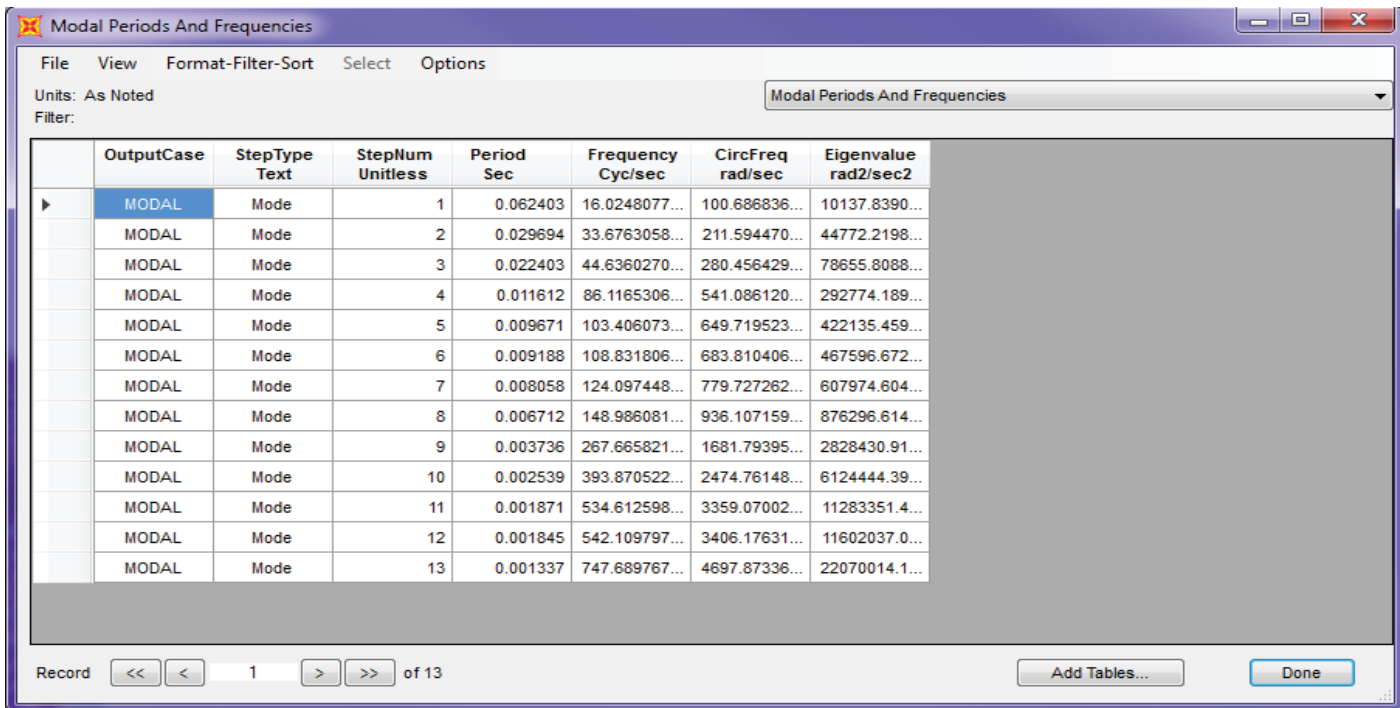
Modal Frequency Output - Lower Bound:

Range of considered operating speeds (+/- 20% per Client Spec):

Vendor published operating Speed = 30 Hz

Design Speed Lower Bound = 30 Hz * 80% = 24 Hz

Design Speed Upper Bound = 30 Hz * 120% = 36 Hz



The screenshot shows a software window titled "Modal Periods And Frequencies". It has a menu bar with "File", "View", "Format-Filter-Sort", "Select", and "Options". Below the menu bar, it says "Units: As Noted" and "Filter:". The main area contains a table with 8 columns: "OutputCase", "StepType", "StepNum", "Period", "Frequency", "CircFreq", and "Eigenvalue". The table lists 13 modes, all of which are "MODAL" and "Mode". The first three modes are highlighted in blue. The bottom of the window shows a "Record" section with navigation buttons and a "Done" button.

OutputCase	StepType	StepNum	Period	Frequency	CircFreq	Eigenvalue
	Text	Unitless	Sec	Cyc/sec	rad/sec	rad2/sec2
MODAL	Mode	1	0.062403	16.0248077...	100.686836...	10137.8390...
MODAL	Mode	2	0.029694	33.6763058...	211.594470...	44772.2198...
MODAL	Mode	3	0.022403	44.6360270...	280.456429...	78655.8088...
MODAL	Mode	4	0.011612	86.1165306...	541.086120...	292774.189...
MODAL	Mode	5	0.009671	103.406073...	649.719523...	422135.459...
MODAL	Mode	6	0.009188	108.831806...	683.810406...	467596.672...
MODAL	Mode	7	0.008058	124.097448...	779.727262...	607974.604...
MODAL	Mode	8	0.006712	148.986081...	936.107159...	876296.614...
MODAL	Mode	9	0.003736	267.665821...	1681.79395...	2828430.91...
MODAL	Mode	10	0.002539	393.870522...	2474.76148...	6124444.39...
MODAL	Mode	11	0.001871	534.612598...	3359.07002...	11283351.4...
MODAL	Mode	12	0.001845	542.109797...	3406.17631...	11602037.0...
MODAL	Mode	13	0.001337	747.689767...	4697.87336...	22070014.1...

Mode 1 (16 Hz) creates a lateral deflection in the X Direction, with the lid deforming as a block.

Mode 2 (33.7 Hz) creates a lateral deflection in the Y Direction, with the lid deforming as a block.

Mode 3 (44.6 Hz) creates a vertical deflection (Z Direction), with the lid deforming as a block.

Modes 4+ create flexible lid deformations, but need not be considered as they only occur at operating speeds
More than double the upper bound of the pump design operating frequency ($2 * 36 \text{ Hz} = 72 \text{ Hz}$)



Calculations and Sketches

DATE: 02-Mar-2017

CONT. NO. A6KV

BY: RPM

CHK'D BY: SC

SHEET NO:

10-P-4400XA/B Series Pump Dynamic Analysis

CALC NO.: PDH-00-ST-CAL-0004

Check Sump / Pump Weight Ratio:

Sump Dimensions:	Outside Diameter =	2.536 m	Inside Diameter =	2.13 m
	Outside Height =	3.653 m	Inside Height =	3.048 m

Approximate Area of Cap Openings =	1.1 m ²
Cap Thickness =	0.3 m

Concrete Volume = 7.261 m³

Concrete Weight = 17426 kg

Pump Weight = 749 kg

Weight Ratio (Sump / Pump): 23.27

The lateral deflection modes will be adequate by inspection based on the Sump / Pump Weight ratio. The sump is so much heavier than the pump that relevant deflections due to the resonant frequencies is unlikely.

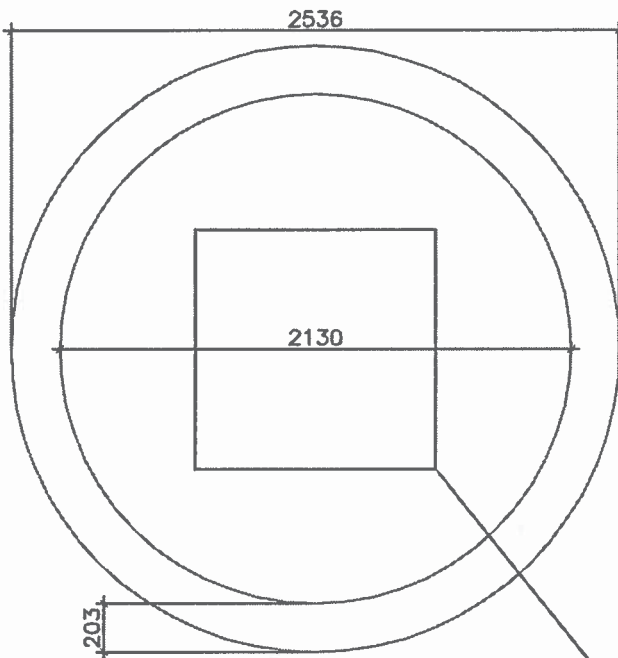
The sump cover is 300mm thick concrete and will therefore be very stiff, meaning that any deflections caused by modal frequency #3 will be very minimal, therefore Mode 3 will be ok by inspection.

Upper Bound:

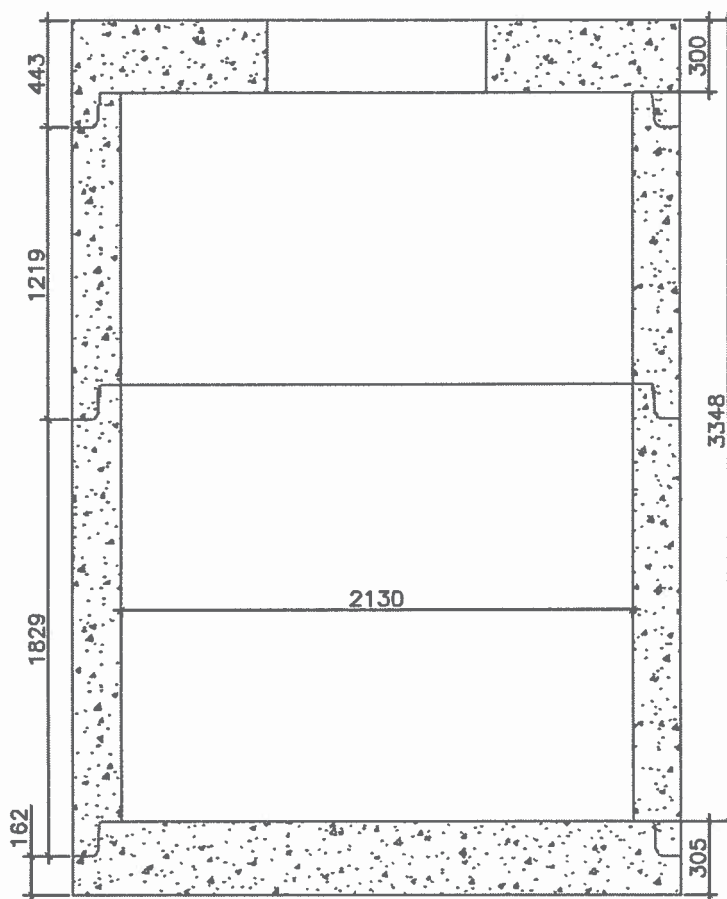
The upper bound will not be investigated as it will be ok by inspection based on the lower bound analysis.

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Appendix A: Vendor Information



ACCESS DETAIL
TO BE DETERMINED



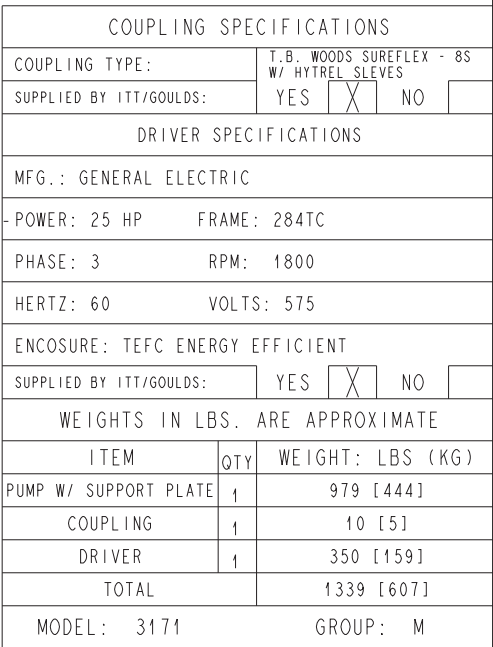
UNITS IN MILLIMETERS UNLESS
OTHERWISE NOTED.

DATE: 2016/09/29		SCALE: SHOWN
FILE: DRAWING1		
DRAWN BY: NC		
CHECKED BY:		
REV:	DATE:	BY:
PRE-POUR		
QC		POST-POUR QC



STORM WATER POND
FLUOR
2100 mm SUMP

ENG. STAMP



CERTIFIED FOR CONSTRUCTION ONLY WHEN SIGNED.

CUSTOMER DATA

GENERAL ARRANGEMENT DRAWING



ANY KIND OF REPRODUCTION ONLY PERMITTED WITH PRIOR WRITTEN CONSENT OF ITT.
ANY RIGHTS IN INTELLECTUAL PROPERTY RIGHTS CONTAINED HEREIN ARE RESERVED BY ITT

THIRD ANGLE

DRAWING

B1-209H368

0

SHEET 1 OF 1

NOTES
- FLANGES CONFORM TO ANSI STANDARDS. BOLT HOLES STRADDLE ϕ. - DIMENSIONAL TOLERANCE TO PIPED CONNECTIONS IS ± 0.50 EXCEPT FOR PUMP SUCTION AND DISCHARGE. - ROUTING OF PIPELINES IS APPROXIMATE AND MAY VARY AFTER ASSEMBLY - REFER TO PUMP IOM FOR TYPICAL GROUTING AND ANCHORING DETAILS.

REV	DESCRIPTION	APPROVAL	DATE

